

ApprovalFlow

Administrator Manual

Configuring forms, approval chains and users, and overseeing requests

This manual is for people with the Admin role. It covers the administrator portal: the dashboard, request oversight and actions, form and approval-chain configuration, reports, user management and the audit log. Server installation is covered separately in docs/DEPLOYMENT-IIS.md. Everyday submitting and approving is covered in the User Manual.

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CHAPTER 1

Overview

Administrators see an extra navigation bar at the top of every page: **Dashboard, Requests, Forms, Lookups, Reports, Users, Audit log** and **Settings**. Everything you see belongs to **your organisation only**: your forms, your people, your requests. Other organisations on the same installation are invisible to you, and you are invisible to them.

Site addresses

Your organisation has **its own address**, and everybody in it - submitters, approvers and administrators - uses that one address. The administrator pages live under **/admin** on it, and only people with the Admin role can open them; everyone else is sent to their home page. Signing in at another organisation address will not work, even with the same email address and key.

Site	Address
User site (give this to submitters and approvers)	http://localhost:8088
Sign-in page	http://localhost:8088/login
Administrator site	http://localhost:8088/admin
Forms and the form builder	http://localhost:8088/admin/forms
Requests	http://localhost:8088/admin/requests
Lookups	http://localhost:8088/admin/lookups
Users	http://localhost:8088/admin/users
Audit log	http://localhost:8088/admin/audit
Settings	http://localhost:8088/admin/settings

NOTE

These are the addresses of the **local trial installation**, which can only be opened on the server itself. In a real installation the first part is your own organisation address, for example <https://acme.approvals.yourprovider.com> or a name of your own such as <https://approvals.acme.com>; the parts after it - `/login`, `/admin` and so on - stay the same. Links in emails always use your organisation address, so an approver who clicks one always lands in the right place.

Help: the manuals in the app

Choose **Help** in the top-right corner, next to your name. Administrators get a small menu with two choices: **Administrator Manual** (this one) and **User Manual** (the one your submitters and approvers see, useful when you explain something to them). Everyone else gets **Help**, which opens the User Manual.

- The manual opens in a new tab, in the browser's own PDF viewer, where it can be searched, saved or printed.
- Only signed-in people can open the manuals, and only administrators the Administrator Manual. The address of an open manual works for a few minutes only - a copied or bookmarked link stops working, so open it again from **Help**.
- The manuals always show the current version: they are updated with the system itself.

Light or dark colours

The small icon next to **Help** in the top bar (and at the top of the sign-in page) switches between three colour choices, one per click:

Icon	Colours
Screen (System)	Follow the setting of your computer or phone: light by day or dark, as you have chosen there. This is where everyone starts.
Sun (Light)	Always light, whatever your computer is set to.
Moon (Dark)	Always dark: black backgrounds with light text and yellow highlights - easier on the eyes in a dim room.

The choice belongs to each person and each browser - it is not an organisation setting, and you cannot set it for others. The FileBank colours (black, yellow and warm white) are the same for every organisation; your **logo** and **name** from Settings appear in the top bar and on the sign-in page.

What administrators are responsible for

- Creating forms, their fields and their approval chains, and keeping approver assignments current.
- Granting roles to people after they register themselves, deactivating leavers, and - rarely - resetting a password key for someone who cannot do it themselves.
- Watching for **overdue** requests and unblocking them (remind, reassign, delegate).
- Answering audit questions using the audit log and its CSV export.

What administrators cannot do - by design

- Approve or reject on someone else's behalf. Only the assigned approver or a delegate can decide. (Assign yourself as delegate if you must act; the audit log records it.)
- Edit a submission or a completed approver section. Completed sections are immutable.
- Reopen a **Rejected**, **Approved** or **Cancelled** request. These states are final; the submitter must submit a new request.
- Change or delete audit log entries.
- See anything belonging to another organisation. Every page and every export is limited to your own, and there is no setting that widens it.

NOTE

An administrator can also be named as an approver in a chain, and can hold the Submitter role. Your home page (click **Approvals** in the top-left) then shows your approval inbox and submissions as for any other user.

CHAPTER 2

Dashboard

The dashboard opens when you choose **Dashboard**. Each tile is a link to the request list with the matching filter applied.

Tile	Counts
In progress	Requests waiting on an approver.
Approved	Requests approved at every step.
Rejected	Requests rejected at some step.
Overdue	In-progress requests whose current step has passed its due date (the step's "Remind after" days, or "Escalate after" if no reminder is set). Steps with neither never become overdue. Highlighted when above zero.
Cancelled	Requests cancelled by an administrator.

Overdue is not a request status: an overdue request is still In progress.

Average time per step

For each form and step: how many decisions have been made and the average time from the step becoming active to the decision. Use it to spot slow steps and to choose sensible reminder periods.

Failed email banner

A red banner appears when emails could not be delivered after 6 attempts. Check the mail server settings with your IT team, then open the affected request and choose **Resend** next to the failed email (see chapter 3).

CHAPTER 3

Requests

Finding requests

The **Requests** page lists every request in your organisation, newest first, 25 per page. Filters combine, and the current filter is part of the page address, so you can bookmark or share it.

Filter	Behaviour
Search	Matches the request number, the submitter's name or email, and the content of the submitted fields. Press Search or Enter.
Form	One form, or all.
Status	In progress, Approved, Rejected, Cancelled, plus the view Overdue .
Submitter	Requests submitted by one person.
Approver	Requests where this person is assigned to, delegated on, or has decided any step. Use it before deactivating someone.
From / To	Submission date range (inclusive).

The request detail page

- **Actions** - the buttons described below, and the archive panel (whether the PDF is stored, and its size).
- **Timeline** - the submission with all its data, then each step: status, who it is assigned to, delegate, waiting-since and due dates, and once decided: who acted, when, from which IP address, the fields they entered and their comments.
- **Emails** - every notification for this request with its delivery status and any error.
- **Audit trail** - every event for this request with user, time and IP address.

Send reminder

Emails the current approver (and delegate, if any) a reminder containing a **fresh link**; their earlier link stops working. Use it when an approver says their link expired. Available only while the request is in progress.

Reassign and delegate

In the Timeline, choose **Reassign...** on any step that is Active or Waiting, pick an approver, and choose **Save**.

Option	Effect
Reassign (box unticked)	The new person replaces the approver. The previous approver's link stops working and they can no longer act. If the step is active, the new approver is emailed a link immediately.
As delegate (box ticked)	The original approver keeps the step and the delegate can act too. Whoever decides first is recorded; the PDF and audit log show who actually acted.

Only active users with the Approver (or Admin) role can be chosen. Reassigning a **Waiting** step changes who will be asked when the request gets there. Completed steps cannot be reassigned.

IMPORTANT

Reassignment changes **this request only**. To change who approves future requests, publish a new version of the form's approval chain (chapter 4).

Cancel request

Choose **Cancel request...**, enter a reason (mandatory) and **Confirm cancel**. The request becomes **Cancelled**, which is final; open steps are closed, approval links stop working and the submitter is emailed your reason. No PDF is produced for cancelled requests. Use this for duplicates or requests submitted by mistake.

Download PDF

Download PDF is available as soon as the PDF exists - usually within a minute of the request closing. It comes from the database. See the chapter **PDF archive**.

Resend a failed email

In the **Emails** table, failed messages show the error and a **Resend** link, which queues the same message again. For approval requests prefer **Send reminder**: the link inside an old message may have been replaced since.

CHAPTER 4

Forms and approval chains

The **Forms** page lists each form with its status, number of steps, chain version, and how many requests it has. A form shown with **none** as its chain needs one before anybody can use it.

Create a form

1. Under **New form**, type a name and choose **Create and configure**. The form is created with one starter field, "Title".
2. Complete the three sections of the editor: **Details**, **Submission fields** and **Approval chain**. Each section has its own save button.

NOTE

Submitters only see a form when it is **Active** and has a **published approval chain**.

Import an existing HTML form

If the form already exists as a web page, you do not have to rebuild it field by field.

1. On the **Forms** page, under **Import an HTML form**, choose **Choose HTML file...** and pick the .html file (up to about 1 MB), or choose **or paste the HTML instead**, paste the source and choose **Read fields**.
2. The **Review the imported form** panel lists every field that was detected. Nothing is saved yet. Correct the name, description, labels, types and **Required** ticks; reorder or remove fields; read the **Worth checking** notes.
3. Choose **Create form and add approval steps**. The form is created and the editor opens.
4. In **Approval chain** choose **+ Add step** for each approver, choose who approves it, try the whole chain in **Preview & test**, and choose **Publish chain**. The form is now available to submitters.

In the HTML	Becomes
Text and search inputs	Text (max length, pattern and placeholder are kept)
Email, tel, url, date, time, datetime-local, month, week, color inputs	The matching control
Number inputs	Number - or Currency when the step is 0.01 or the label mentions amount, cost, price, total... Min and max are kept.
Range inputs	Slider, with its min and max
Text area	textarea
Drop-down (select)	select with the same options; an empty "choose..." option is dropped
Radio buttons sharing a name	Radio buttons; the group label comes from the fieldset legend
Single tick box	checkbox
Several tick boxes sharing a name, or a multi-select	Tick boxes (many)
File upload, password, hidden, disabled fields and buttons	skipped - file uploads and passwords are listed under Worth checking

- Labels are taken from the matching label element, a wrapping label, aria-label or placeholder - in that order - and otherwise from the field name.
- The file is only read for its fields. Scripts, styling, the form's action address and any JavaScript validation are ignored; drop-downs that a script fills in arrive empty and are imported as text.
- Importing the same file twice creates a second form; the name must be different because it determines the form's internal address.

Sample forms to start from

The demo organisation comes with five ready-made forms, each with a published approval chain. Use them to try the system, to show people how it works, or as a starting point: open one, change what you need in **Submission fields** and **Approval chain**, and try it in **Preview & test** before anybody uses it. Your own organisation can have them too - ask your provider.

Form	What it shows	Approval chain
Leave Request	A department lookup that fills in the department code; an employee number that must be 4 to 8 digits; calendar days worked out from the first and last day; a typed signature.	Line manager, with reminders and escalation to a director
Mileage Reimbursement	A table of trips whose mileage is worked out at \$0.70 a mile, with column totals; a total claim worked out from the table; a required "I certify" tick box; a drawn signature.	Manager, then Finance
IT Access & Equipment Request	Choices with tick lists (access and equipment), a school lookup, a required reason of at least 10 characters.	The submitter picks their manager , then IT Services
Records Retrieval Request	A table of boxes and files, each with a delivery choice (scan, deliver, pick up); urgency; a needed-by date.	Records manager, escalating after 2 days
Secure Destruction Authorization	A table of boxes with retention dates and a cubic-feet total; the destruction method; a required "no litigation hold" confirmation; an optional certificate of destruction; a drawn signature.	Department head, then records officer

TIP

Sample forms are ordinary forms: requests made on them are real requests, emailed and archived like any other. Before letting people use one for real, check its approvers, and switch off (untick **Active**) or delete any sample you do not need.

Delete a form

On the **Forms** page choose **Delete** in the form's row. A confirmation explains what will happen for that particular form; choose **Delete form** to go ahead.

The form has...	What Delete does
No requests	Deleted permanently , together with its fields and every version of its approval chain.
One or more requests	Removed from the repository. It disappears from this list and from submitters, and cannot be started, edited or reactivated. Its requests stay in Requests , with their timeline, audit trail and PDFs; requests still in progress carry on to a decision with the approvers they already have.

- Either way the name becomes free, so you can create or import a new form with the same name.

- Deleting cannot be undone from the portal. To pause a form temporarily, open it and untick **Active** instead - that stops new submissions and can be reversed at any time.
- The deletion is recorded in the audit log as **form.deleted** with your name and what was kept.

Details

- **Name** and **Description** - shown to submitters on their home page.
- **Active** - untick to stop new submissions. Requests already in progress continue normally.
- **Submitters can see approver comments and fields in the portal** - off by default. When off, submitters see only each step's decision, approver name and date (and always the rejection reason). Applies immediately to existing requests too.

IMPORTANT

The archived PDF always contains the full record, including approver comments, and submitters can download the PDF of their own request. If approver comments must stay confidential for a form, raise this with your system owner before go-live.

Submission fields: the form builder

The builder shows a **live preview** of the form on the left - laid out exactly as submitters will see it - and a side panel on the right.

To...	Do this
Add a control	With nothing selected, the side panel shows Add a control . Click one (for example + Phone) to add it at the end, or drag it onto the preview to drop it exactly where you want it.
Move a control	Drag it to another place in the preview. A blue bar shows where it will land: before or after the control under the pointer (left or right of a control that shares its row, above or below a full-width one). Controls always stay in rows on the grid, so there is no free positioning.
Edit a control	Click it in the preview. The side panel becomes its properties ; every change shows immediately. Choose Done , or click an empty part of the preview, to go back to the list of controls.
Resize a control	Drag the bar on its right edge . It snaps to a quarter, a third, half, two thirds, three quarters or full width, so neighbouring controls line up in rows. Or use the Width buttons in the properties.
Change a text area's height	Drag the bar on its bottom edge , or set Height (rows) .
Reorder, copy, delete	Select the control and use the small toolbar at its top right: up, down, duplicate, delete. Up and down also work from the keyboard.
Save	Save fields . Nothing changes for submitters until you save. Problems that would block the save (a missing label, a duplicate key, a choice control without options) are listed under the builder.

On a phone-sized screen the form always collapses to a single column, whatever widths you set.

Controls

Control	Notes
Text, Long text	Up to 500 / 4,000 characters. Optional minimum and maximum length; Text can also have a pattern.
Email, Phone, Web address	Checked for the right shape. Web addresses must start with http:// or https:// and are shown as links.
Number, Currency	Optional minimum, maximum and step. Currency is stored with two decimals.
Slider	A number chosen by sliding, 0 to 100 unless you set minimum and maximum.
Date, Time, Date & time, Month, Week	The browser's own pickers.
Drop-down, Radio buttons	One choice from your list of options (one per line). Radio buttons can be shown side by side.
Tick boxes (many)	Any number of choices from your list; optional minimum and maximum number ticked.
Single tick box	Yes/No. If required, it must be ticked - use it for "I confirm..." statements.
Lookup (Excel)	A drop-down whose choices come from an imported Excel table; other controls can be filled in from the chosen row. See the chapter Lookup tables .
Colour	A colour picker; stored as a code such as #1d4ed8.
Signature (draw)	A signing box: the person signs with a finger, a stylus or the mouse, with Undo and Clear. The drawing appears in the approver's view, on the request page and in the PDF. It cannot be pre-filled or copied from an earlier request. The system records their account, the time and their IP address.
Signature (typed name)	The person types their full name; the system records their account, the time and their IP address. Use this one if people may need to sign with a keyboard only.
Heading, Paragraph, Divider	Layout only: they organise the form and are never submitted, stored or printed.
Picture	Layout only: a picture such as your logo, shown on the form. Upload a PNG, JPG, GIF or WebP file of up to 300 KB, set its largest height in pixels and a short description for screen readers. It is not printed on the archived PDF.

Properties

- **Label** - what people see. **Required** - must be filled in (an approver can still reject without completing required fields).
- **Control type** - can be changed later; existing requests keep the type they were submitted with.

- **Placeholder** - grey example text inside an empty box. **Help text** - a line of guidance under the control.
- **Width** - headings, paragraphs and pictures can be resized like any other control (Width buttons, or drag the right edge), so a logo and a heading can sit side by side; a divider always runs the full width. **Size** (headings only) - Small, Medium (the default), Large or Extra large. **Alignment** - headings and paragraphs can be left, centre, right or justified; a picture left, centre or right. For boxes people type into (text, numbers, currency, email and similar) **Text alignment** lines up what is typed inside the box - right is useful for amounts. A long-text box can also be justified.
- **Default value** - pre-filled for new requests. For a single tick box type yes; for tick boxes (many) list the options separated by commas; for dates use YYYY-MM-DD.
- **Advanced > Key** - the control's internal name, filled in from the label. Do not change the key of a control that is already in use: to the system that is a different field.
- Removing a control **retires** it. Existing requests keep the data and labels they were submitted with; only new submissions change. The same is true of renamed labels.

NOTE

File uploads are not available: requests and their PDFs carry form data only.

Calculated fields

A **Number**, **Currency** or **Text** control can be worked out from other controls instead of typed in - a subtotal, a tax amount, hours worked, a full name, a reference code. Select it, tick **Work this out from other fields** under **Calculated**, and type a formula using the other controls' keys. The properties list the keys you can use, and **Functions** lists every function.

Formula	Gives
quantity * unitPrice	Quantity times unit price.
items.amount + delivery	The total of the amount column of the data grid items , plus delivery.
subtotal * taxRate / 100	A percentage of another calculated control.
elapsed(start, end)	The hours between two times, also across midnight (22:00 to 06:30 is 8.5).
round(total / 12, 2)	A monthly amount, rounded to 2 places.
total - discount	Subtraction is the - sign; + - * / and brackets work as usual.
firstName & " " & upper(lastName)	Joins text: "Ada LOVELACE". Text in a formula goes in double quotes.
substring(code, 5, 4)	Four characters starting at the 5th: from INV-2026-0042 gives 2026.
if(amount > 1000, "Director", "Manager")	Chooses between two results. Compare with = <> < > <= >=.
days(from, to) + 1	The number of days a leave request covers, both days included.

Functions	
Numbers	sum(a, b, ...), average(a, b, ...), min, max, round(x, places), floor(x), ceiling(x), abs(x), mod(a, b), power(x, n)
Dates and times	days(from, to) days between two dates; elapsed(start, end) hours between two times
Text	concat(a, b, ...) or &, substring(text, start, length) (also called mid), left(text, n), right(text, n), upper, lower, trim, len(text), replace(text, find, with), text(number, places)
Logic	if(condition, then, otherwise)

- A formula can use any control that holds one value - text, numbers, dates, drop-downs, lookups, tick boxes (1 when ticked, 0 when not), times (as hours) - other calculated controls in any order, and a data grid column's total written **grid.column**. Drawn signatures, tick boxes (many) and data grids themselves cannot be used.
- On the form the box is read-only, labelled **calculated**, and updates as people type. When the form is submitted the server works it out again, so nobody can type in a result of their own.
- An empty control counts as 0, but if every control a formula uses is empty the result stays empty. A missing time is never counted as midnight.
- Currency results have 2 decimals; for a number choose **Decimal places** (0 to 6). **Minimum** and **Maximum** apply to the result - for example a total that may not exceed 10,000.
- The form cannot be saved while a formula uses a key that does not exist, uses itself, or two controls depend on each other; the problem is listed under the builder. Approver sections can have calculated controls too, from that section's own controls.

Approval chain

Steps run strictly in order, one approver per step. For each step set:

Setting	Meaning
Step name	Shown to everyone, e.g. "Line manager", "Finance".
Who approves this step	Always the same person (an active user with the Approver or Admin role), or Chosen by the submitter (step 1) or the approver of the step before (later steps) - see below.
Remind after (days)	First automatic reminder. Also the point at which the step counts as overdue.
Then repeat every (days)	Further reminders. Leave empty for a single reminder. At most 10 reminders are sent per step.
Escalate after (days)	One escalation notice if the step is still undecided.
Escalate to	Who receives the escalation. "Administrators" sends it to the organisation's admin notification address, or to all active admins.

Who approves each step

Every step has a **Who approves this step** box at the top. Choose one of:

Choice	What happens
Always the same person	Pick the approver. The step always goes to them.
Chosen by the submitter (step 1)	The person filling in the form chooses who approves step 1, in a Send to box at the bottom of the form, before submitting.
Chosen by the approver of step N (later steps)	The approver of the step before chooses who approves this one, in a Send to box next to Approve . Each step can use its own list.

For a chosen step, set **From**:

- **A lookup file** - an Excel list of the people who may approve this step, imported on the **Lookups** page: one row each, with a column for their **sign-in email address** and any other columns you like (school, name, phone). Choose the file, the **Column with their email** and optionally the **Column with their name** (both are guessed from the headings), and tick any other columns to **show, filled in, beside the chosen person**. Each step can use a different file - for example a list of secretaries for step 1, of principals for step 2 and of payroll managers for step 3.
- **Everyone with the Approver role** - the chooser picks from the approvers on the Users page.

The person choosing sees the file's key column (for example the school) with the name beside it - "Hawes School - Maria Monnerat" - and must pick one: a chosen step has no default. Choose **Publish chain** to save.

- **The file decides who can be chosen, not the Users page.** When someone **without an account** is chosen, an account is created for them at that moment, with the Approver role and the name from the file. They receive the usual email with the link; when they open it they confirm their email address with a code and choose their 6-digit password key, like anyone signing in for the first time. Someone who has an account without the Approver role is given the role. Both are in the audit log (**user.created_from_list**, **user.role_granted**).
- Nobody can choose themselves, and an approver cannot choose the person who submitted the request. Someone you have **deactivated** cannot be chosen, and a row without a usable email address is not offered.
- A request can always be **rejected** without choosing anybody.
- Until the step before it is decided, a chosen step shows **(chosen by the submitter)** or **(chosen by the previous approver)** instead of a name.
- Each choice is recorded as **step.approver_chosen**, with the file and the row. You can still **Reassign** a step on the request page afterwards.
- A file a step chooses from cannot be deleted, and **Replace data** refuses a file without the columns the step uses.

IMPORTANT

Because being in the file is what allows a person to approve, treat the file like a permission list: only administrators can import or replace lookup files, and every import is recorded in the audit log.

Approvers do not fill in controls of their own: each approver sees the submitted form and gives a **decision**, a **comment** and, when rejecting, a **reason**. Requests from before this change keep and print what their approvers filled in.

Use the arrows to reorder steps, the cross to remove one and **+ Add step** to append. Nothing takes effect until you choose **Publish chain**.

IMPORTANT

Publishing creates a **new version**. Requests already in progress keep the version they started with - their steps and approvers do not change. To change the approver on an in-flight request, use **Reassign** on that request.

Escalation only **notifies**; it never reassigns. The recipient is given a link to the request so they can remind, reassign or delegate.

Preview and test the chain

The **Approval chain** section has two modes at the top, like the form builder: **Design** and **Preview & test**. Preview walks a pretend request through every stage - the submission and each step - showing what that person sees. It uses the chain and the submission fields **as they are on your screen, including changes you have not saved or published**. Each step also has its own **Preview & test** button, which opens the preview at that step.

1. Choose **Preview & test**. A row of stages appears: **Submission**, then each step by number and name, then **Approved** (or **Rejected**). Click a stage to jump to it. A step whose settings are incomplete has a dashed outline.
2. In **Submitted by**, choose who the pretend request comes from - you, by default. It matters because nobody can choose themselves or the submitter.
3. **Submission**: the form appears as the submitter gets it, with the **Send to** box for step 1. Fill it in, pick the step 1 approver if the step is chosen, and choose **Test submit**. The answers are checked exactly as for a real request, then the preview moves on to step 1.
4. **Each step**: you see the approver's screen - the step and who receives it, the submitted answers, the earlier decisions, and the step's reminder and escalation timetable in words. Add a comment, pick who approves the next step if it is chosen, and choose **Test approve and send on**. Or choose **Test reject...**, give a reason and choose **Test rejection**.
5. At the end a summary lists who submitted, who decided each step and with what comment, and what the submitter would have been emailed.
6. Choose **Edit step N in Design** (or **Back to Design**) to fix what you found, or **Start again** to clear the pretend request.

The people offered in each **Send to** box are worked out by the server, exactly as for a real request: from the step's lookup file, or from everyone with the Approver role, leaving out the person choosing and the submitter. If a step cannot be handed on, the preview says why - for example an approver who has been deactivated or has lost the Approver role, a lookup file that has been deleted, or a column the file no longer has. These are the same problems that would stop **Publish chain**.

TIP

A chain preview saves nothing: no request is created, nobody is emailed, nothing appears in the audit log, and choosing someone from a lookup file does **not** create their account. You can preview a chain before it has ever been published.

File names

Approved: **[FormName]_[RequestNumber]_[YYYYMMDD].pdf**. Rejected: **[FormName]_[RequestNumber]_REJECTED_[YYYYMMDD].pdf**. Spaces in the form name become hyphens and characters that are not allowed in file names are removed; the date is the UTC date the request closed. Example: Purchase-Request_REQ-000042_REJECTED_20260920.pdf.

CHAPTER 5

Data grids (tables with calculated columns)

A **Data grid** is a control that holds a table: line items on a purchase request, journeys on an expense claim, people on a training booking. The person filling in the form **adds as many rows as they need**, and columns can be **calculated** from the others - quantity times unit price, for example - with a **total** underneath.

Add a data grid to a form

1. In the form builder choose **+ Data grid** (under *Other*). It starts as a line-items table: Description, Quantity, Unit price and a calculated Amount with a total.
2. Click the grid and open a column under **Columns** to change its **Heading** and **Type**: Text, Number, Currency, Date, Time, Drop-down (give it options) or **Calculated**.
3. For a Calculated column type a **Formula** and choose the **Decimal places**.
4. Tick **Required in every row** for cells that must not be left empty, and **Show a total under this column** for number, currency and calculated columns.
5. Use the arrows to reorder columns, **+ Add column** for more (up to 12), and **Min. rows / Max. rows** to limit how many rows can be entered (at most 200).

Formulas

A formula uses the **key** of other columns (shown next to each column's type, and listed under the formula box) with **+** **-** ***** **/** and brackets. It can read number, currency and time columns, and calculated columns to its left. A **time counts as hours** (17:30 is 17.5), so subtracting two times gives the hours between them.

Formula	Meaning
quantity * unitPrice	Line amount
timeOut - timeIn	Hours between two Time columns: 09:00 to 17:30 gives 8.50. A shift past midnight comes out negative (22:00 to 06:00 gives -16.00): use elapsed() for timesheets
timeOut - timeIn - breakHours	Hours worked, less a break typed as a number (0.5 = half an hour)
elapsed(timeIn, timeOut)	Recommended for timesheets. The same, but a time out before the time in means the next day: 22:00 to 06:00 gives 8.00
elapsed(timeIn, timeOut) - breakHours	Hours worked across midnight, less a break
hours * rate	Pay, from a calculated hours column
amount * 0.13	13% tax on a calculated column called amount
round(hours * rate, 0)	Rounded to whole numbers
max(nights - 1, 0) * allowance	Never below zero
(km - min(km, 50)) * 0.45	Only the distance above 50 km

- Available functions: **round(x, places)**, **min(a, b, ...)**, **max(a, b, ...)**, **abs(x)**, **elapsed(start, end)**.
- An empty cell counts as 0. If every cell a formula reads is empty, or the result is not a number (division by zero), the calculated cell stays empty. A missing **time** never counts as 0: the cell stays empty until both times are entered.
- The builder tells you at once if a formula cannot be read or uses a key that is not a number column. Try it in **Preview & test** before publishing.

NOTE

Calculated cells and totals update on screen while the person types, but the **server works them out again** when the form is submitted and stores its own results. Nobody can submit a different amount than the formula gives. Rows left completely empty are ignored. The table, with its totals, appears in the approver's view, the request page and the PDF.

CHAPTER 6

Lookup tables (Excel)

A lookup table is reference data you keep in Excel - schools and their secretaries, departments and their codes, job titles and their accounts. Import it once, and any form can offer its key values in a drop-down and **fill in related boxes automatically**.

Prepare the spreadsheet

- Save it as an **Excel workbook (.xlsx)**. Older .xls and .csv files: open them in Excel and use Save As.
- The **first row** holds the column headings, for example *School, Department, Secretary, Email*. Headings must all be different.
- One row per item. One column is the **key** - the value people choose on the form; it must be filled in on every row and never repeat.
- The first visible sheet is used. Formulas are imported as their calculated values, dates as YYYY-MM-DD. Limits: 5,000 rows, 30 columns, 5 MB.

Import

1. Open **Lookups** and choose **Choose .xlsx file...**
2. Check the preview: sheet name, number of rows, the first rows. Nothing is saved yet.
3. Give the table a **name** and choose the **key column**, then choose **Import lookup table**.

If the file cannot be used, the message says why - for example a repeated or missing key value, two columns with the same heading, or a file that is not .xlsx.

Use it on a form

1. In the form builder choose **+ Lookup (Excel)**, give it a label such as "School", and pick the **Lookup table** in its properties.
 2. Add (or select) a Text, Long text, Email, Phone, Web address, Number or Currency control for each related value - for example "Secretary".
 3. In that control's properties, under **Auto-fill from a lookup**, set **Follows** to the lookup control and **Shows column** to the column to display. The preview marks it *auto-filled*.
 4. Save the fields. On the form the box is read-only and fills in as soon as a choice is made; changing the choice updates it, clearing the choice empties it.
- A form can use several lookup tables, and several controls can follow the same lookup. Approver steps can use lookups too.
 - Tables with more than 60 entries become a type-to-search box instead of a drop-down.
 - **Auto-filled values are worked out by the server** when the form is submitted, from the table as it is at that moment. Whatever a browser sends for those boxes is ignored, so they cannot be tampered with.
 - Only the key values and the columns a form actually uses are sent to the people filling it in. Other columns of the spreadsheet stay on the server.

- The values are stored with the request like any other answer: they appear in the approver's view, the timeline and the PDF, and **never change afterwards**, even if the table does.

Test the auto-fill before publishing

Every form builder has two modes at the top: **Design** and **Preview & test**. Preview works on the form as it is on your screen, **including changes you have not saved**, and also on forms that are inactive or have no approval steps yet.

1. Choose **Preview & test**. The form appears exactly as the person filling it in will get it.
2. Make a choice in a lookup control. The boxes that follow it fill in at once and are read-only; change the choice and they change, clear it and they empty.
3. Fill in the rest and choose **Test submit**. The server checks the answers with the same rules as a real request. Problems appear next to the boxes; otherwise you see the list of values that would be stored, with the auto-filled ones marked.
4. Choose **Design** to go back and adjust, or **Start again** to clear the answers.

NOTE

A test submit saves nothing: no request is created, nobody is emailed and nothing appears in the audit log. If Preview lists problems instead of the form (a lookup control without a table, an auto-filled box without a column), fix them in Design first.

Keep the data up to date

On the **Lookups** page choose **Replace data** and pick the new file. The key column stays the same. New requests use the new data at once; requests already submitted keep their values. The file must still contain every column that forms auto-fill from - otherwise the message lists the missing columns, and nothing is changed.

Click a table's name to see its first 100 rows. **Delete** removes a table that no form uses; the **Used by** column shows which forms still use it.

Change individual rows

Click a table's name to open it. You can change it one row at a time, without a new spreadsheet:

To...	Do this
Find a row	Type any part of any cell in Find a row and choose Find . Only the first 100 rows are listed, so use Find to reach the others. Show all clears the search.
Change a row	Choose Edit on the row; it turns into boxes. Change the values and choose Save (or press Enter). Cancel or Escape leaves it as it was. The row keeps its place in the table.
Delete a row	Choose Delete on the row, then Yes, delete .
Add a row	Fill in Add a row under the rows and choose Add row . The row goes at the end of the table.

The key column must be filled in and must differ from every other row (capitals do not count as a difference); the other columns may be left empty. Forms use the change at once: a deleted row is no longer offered, and auto-filled boxes show a changed row's new values. Requests already submitted keep the values they were given. Each change is recorded in the audit log as **lookup.row_added**, **lookup.row_updated** (with the values before and after) or **lookup.row_deleted**.

NOTE

Changes made here are not in your spreadsheet. **Replace data** puts the whole table back to what the new file contains, so make the same changes in the spreadsheet before you next replace the data, or they will be lost.

NOTE

If someone's saved choice disappears from a replaced table (a school closes), existing requests are unaffected; on new requests that value is simply no longer offered.

CHAPTER 7

Users

You do not create users

People create their own accounts. The first time someone signs in with their work email, they receive a verification code by email and choose their own **6-digit password key**. They then appear on the **Users** page with the **Submitter** role and the date they joined. You never see, set or send anyone's key.

- If the server is configured with allowed email domains, only addresses at those domains can register.
- The **first administrator** is created once, when the system is installed (see the deployment guide). Further administrators are ordinary users you give the Admin role to.

Role	Gives access to
Submitter	Submitting forms and viewing their own requests. Given automatically at registration.
Approver	Deciding steps assigned or delegated to them. Required before someone can be chosen in a chain or reassignment.
Admin	The administrator portal. Admins can also be chosen as approvers.

Making someone an approver

1. Ask the person to sign in once, so that their account exists.
2. On the **Users** page tick **Approver** in their row. The change is saved immediately.
3. They can now be selected in an approval chain (Forms) or when reassigning a step.

IMPORTANT

Someone who has never signed in cannot be chosen as an approver. When you plan a new form, ask its approvers to sign in first.

Change roles

Tick or untick the role boxes in the user's row; the change takes effect immediately, including for people who are signed in. A user must keep at least one role, and you cannot remove your own Admin role.

Status column

Status	Meaning
Active	Has a password key and can sign in.
No key yet	You reset their key and they have not created a new one yet.
Locked	Five wrong keys in a row. Unlocks by itself after 15 minutes; a reset also clears it.
Inactive	Deactivated by an administrator. Cannot sign in or register again with that address.

Forgotten password keys

People normally fix this themselves: on the sign-in page they choose **Forgot your key?**, receive a verification code by email and choose a new key. You are only needed when that route is not available to them - for example they cannot receive the email - or when you want to force someone to choose a new key. In that case choose **Reset password key** in the user's row and confirm.

- Their old key stops working at once and they are signed out on every device.
- The next time they sign in with their email, they receive a verification code and create a new 6-digit key - exactly like a first sign-in. Roles, requests and approvals are kept.
- Approval steps assigned to them stay assigned; their emailed approval links work again once they have signed in.
- The reset is recorded in the audit log as **user.key_reset** with your name.

IMPORTANT

Be sure the request really comes from that person (call them back, or ask in person). Because the new key can only be created by someone who can read that person's mailbox, a reset requested by an impostor does not give them access - but it does sign the real user out.

Deactivate and reactivate

Deactivate signs the person out everywhere and blocks sign-in at once. Their history is kept. If they still have approval steps assigned, a message tells you how many: open **Requests**, use the **Approver** filter and reassign those steps; then publish a new chain version for any form that names them. **Reactivate** restores access with the same roles and key. You cannot deactivate yourself.

IMPORTANT

While a form's current chain names an inactive approver, new submissions of that form are refused with a message asking the submitter to contact an administrator. Update the chain first when someone leaves.

CHAPTER 8

Reports

The **Reports** page turns what people entered on a form - in the submission, in every approver's section and in data grids - into a table you can read on screen, export to **Excel** or **CSV**, and save to run again. Only administrators can use it, and every report only ever shows your own organisation.

Building a report

1. Under **New report on the form**, choose the form. A report is always about one form.
2. Choose **One row for each: Request**, or - if the form has a data grid - **Line of the data grid**, which lists every line (every purchase line, every timesheet row) with its request's details beside it.
3. Under **Columns**, add what you want to see: the request itself (number, status, who submitted it, dates, the step it is waiting at, days to decide), any control of the form, any control of an approver's section, and the total of a data grid's number column. Use the arrows to order them.
4. Under **Filters**, narrow it down: the submitted dates, the status, and conditions on any column - is, is not, contains, more / less than, between, before / after, is empty, is filled in. Text is compared ignoring capitals.
5. Choose **Run report**.

The result lists the newest requests first (or in the order you choose under **Sort by**) and totals every number column. Up to 5,000 rows are shown on screen; the exports always contain every row.

Grouping and summaries

Choose a column under **Group by** to get one row per value - per department, per school, per status - with how many requests (or lines) each has. For each other column choose how it is summarised:

Total, **Average**, **Lowest** or **Highest** for numbers, **Count** for anything else. Grouping by a date asks whether to group **per day, week, month or year**; the groups are labelled 09/23/2026, Week 39, 2026, 09/2026 and 2026, in date order. A grouped report also shows a bar chart of its first summary.

Example	How
Spend per department this year	Columns: Department, Amount. Filter: submitted from 1 January. Group by Department.
Hours per school per month	One row for each line of the timesheet grid. Columns: School, Worked hours. Group by the date column, per month. (For hours per person and day, use the Hours report below.)
Requests still waiting and where	Filter: status In progress. Group by Waiting at step.
Every purchase over 1,000	Columns of your choice. Filter: Amount is more than 1000.

Exporting and saving

- **Export to Excel** gives a workbook with real numbers and dates (shown as MM/DD/YYYY, so they still sort and filter as dates), a bold header and a totals row. **Export to CSV** gives a plain file that any program can open. Every export is recorded in the audit log.
- To keep a report, type a **name** and choose **Save report**. Saved reports are listed at the top of the page for every administrator; open one to run it again with today's data. Change it and choose **Save changes**, or **Save as a new report** to keep both.
- Reports run when you open or run them - nothing is sent automatically.

TIP

Reports show dates as **MM/DD/YYYY** (for example 09/23/2026) and a date with a time as **MM/DD/YYYY h:mm AM/PM** (09/23/2026 2:05 PM) - on screen, in Excel and in CSV alike. Times are shown in the time zone of the computer you run or export the report on. The boxes where you choose filter dates follow your browser language, which for US English is also MM/DD/YYYY.

NOTE

Renamed controls keep their history: a report goes by each control's key, so old and new requests line up. Controls removed from the form still appear, marked (retired). One run reads up to the newest 20,000 requests; narrow the dates or status if the page says it stopped there.

Hours report (timesheets)

At the top of the Reports page, switch from **Report builder** to **Hours**. It lists the lines of a timesheet form - **Date, School, Time In, Time Out and Worked Hour** - for one person or for everyone, by the **day worked** (the date on each line, not the day the timesheet was submitted).

1. Choose the **Form**. Only forms with a timesheet grid are offered: a data grid with a date column and Time In / Time Out (or Worked Hour) columns. With a single timesheet form it is chosen for you.
2. Choose the **Submitter**, or leave **All submitters**. The list holds everyone who has submitted that form.
3. Choose **Single day** and the day, or **Date range** and the **From** and **To** days. **This week, Last week, This month and Last month** fill in the range (weeks start on Monday).
4. Only **approved** timesheets count. Tick **Also count timesheets still waiting for approval** to include those too. Rejected and cancelled timesheets never count.
5. Choose **Run report**, then **Export to Excel** or **Export to CSV** if you need a file.

Chosen	The result
One submitter	Their lines in date order, with the total hours and number of days at the bottom.
All submitters	A Submitter column, the lines grouped by person with a subtotal after each (hours and days), and the total of everyone at the end.

- Hours are shown with 2 decimals (7.50 is seven and a half hours). Each line also names its request, so you can find the timesheet it came from.
- Worked Hour is the value stored on the timesheet. Timesheet forms should work it out with **elapsed(timeIn, timeOut)** (see **Formulas** in the chapter **Data grids**), so that a shift past midnight - 22:00 to 06:00 - is stored as 8 hours. Timesheets submitted under an older **timeOut - timeIn** formula stored such a shift as a negative number; the report ignores a missing or negative value and works the hours out from Time In and Time Out instead.
- Which column is which is guessed from the headings. If a form uses other headings, open **Columns used from ...** and choose them; your choice and the last dates you used are remembered in this browser.
- Dates are written **MM/DD/YYYY** on screen, in Excel and in CSV; the file name uses the same dates with underscores (Hours_-_All_submitters_-_09_01_2026_to_09_05_2026.xlsx), because a file name cannot contain slashes. The Excel file has a title line with the form, person and dates, real dates and numbers, and bold subtotal and total rows. Every export is recorded in the audit log as **report.exported**.

CHAPTER 9

Audit log

Every state change is recorded with the **user**, **UTC time** and **IP address**: sign-ins and failures, submissions, step activations and decisions, reassignments, reminders, escalations, cancellations, configuration and user changes, PDF generation, uploads, downloads and exports. Entries can never be edited or deleted, by anyone.

Browsing

Filter by **Event contains** (for example "step." or "auth.login"), **User** and date range. Hover over a row to see its detail data. Click a request number to open that request. Newest entries are first, 50 per page.

Export to CSV

Export CSV downloads everything matching the current filters (not just the visible page), including user email, browser and the full detail column. The export is itself logged. Values that a spreadsheet could interpret as a formula are prefixed with an apostrophe.

Event reference

Event	Meaning
auth.login / auth.login_failed / auth.logout	Sign-in activity.
auth.account_locked / auth.login_blocked_locked	Fifth failure locked the account / attempt while locked.
auth.refresh_reuse_detected	An old session token was replayed; all of that user's sessions were ended. Investigate if unexpected.
auth.account_created / auth.key_created	Someone registered at first sign-in / created a new key after a reset. Detail shows whether the email was verified.
auth.password_changed	A user changed their own password key.
auth.key_reset_requested / auth.key_reset_self	A user asked for a forgotten-key code / chose a new key with it.
request.submitted / approved / rejected / cancelled	Request lifecycle, with from and to states.
step.activated / approved / rejected	Step lifecycle. Detail shows asDelegate, viaLink and selfApproval when they apply.
step.reassigned / step.delegated	Administrator changed who can act.
step.reminder_sent / step.escalated	Manual or scheduled reminder; escalation (user "System" when automatic).
approval.link_opened	An approver opened their emailed link.

Event	Meaning
archive.pdf_stored / pdf_copied_to_database	The PDF of a closed request was stored in the database (older entries may also show SharePoint uploads, from when that was supported).
pdf.downloaded	Somebody downloaded a request's PDF.
report.exported / saved / updated / deleted	Reports exported to Excel or CSV, and saved reports changed.
pdf.downloaded / audit.exported	Someone downloaded a PDF / exported the log.
lookup.imported / replaced / deleted	An Excel lookup table was imported, had its data replaced, or was deleted.
form.created / updated / fields_updated / deleted, form.import_parsed, chain.published	Configuration changes (form.deleted shows whether it was removed outright or kept for its requests).
user.updated / user.key_reset	An administrator changed roles or status / reset a password key.
notification.failed / requeued	An email was given up on / queued again by an admin.
settings.updated	An organisation setting was changed (chapter 11). The detail lists which settings, never their values, and names a provider administrator if one made the change.
user.impersonation_started	A provider global administrator started a support session as one of your administrators. The detail shows who and why.
tenant.created / updated / suspended / reactivated	Your organisation was set up, renamed or moved to another address, or its access was suspended and restored, by a provider global administrator.

selfApproval marks a step where the approver is also the request's submitter. It is allowed, and flagged so it can be reviewed.

impersonatedByPlatformAdmin in the detail of an entry means it was done during a provider support session rather than by the named person themselves (chapter 11).

CHAPTER 10

PDF archive

When a request is approved or rejected, a PDF of the complete record is made and **kept in the database**. From there it can be downloaded from the request page by administrators, by the submitter, and by the approvers of that request. A stored PDF can never be changed or deleted - not even by an administrator - so it stays a faithful record of what was decided.

The PDF is made **after** the decision is saved and the submitter's email is queued, so it never delays notifications. There is nothing to set up: every organisation's PDFs are kept this way. To take PDFs elsewhere, download them from the request page, or list requests on the **Reports** page.

Archive status	Meaning
PdfPending	Request just closed; the PDF will be made within about 15 seconds.
Stored	The PDF is in the database and can be downloaded.

CHAPTER 11

Notifications, reminders and escalation

Email	To	Sent when
Submission received	Submitter	A request is submitted.
Approval needed	Approver	Their step becomes active.
Approval needed (reassigned)	New approver or delegate	An admin reassigns or delegates an active step.
Reminder	Approver and delegate	Admin chooses Send reminder, or the reminder rule is due.
Escalation	Escalation user or admins	The step passes its "Escalate after" days. Once per step.
Rejected	Submitter	Any step is rejected. Includes reason, step, approver, date.
Rejected (admin alert)	Administrators	Same moment as above.
Fully approved	Submitter	The final step is approved.
Cancelled	Submitter	An admin cancels the request.
Verification code	The person signing in	First sign-in, "Forgot your key?", or first sign-in after an administrator reset.

"Administrators" means the organisation's admin notification address if one was set when the organisation was created, otherwise every active user with the Admin role.

Emails are queued with the change that caused them and sent within about 10 seconds. A failed send is retried after 1, 5, 15, 60 and 180 minutes; after the sixth failure it is marked Failed and shown on the dashboard.

Reminder and escalation rules are checked every 5 minutes. Days are calendar days counted from when the step became active (reminders repeat from the previous reminder). Rules come from the chain version the request started with.

CHAPTER 12

Settings

The **Settings** page holds the choices that apply to your whole organisation: how the portal looks, who may register, what your emails are sent as, and where approved and rejected PDFs are archived. Each setting can be left blank, which means **use the installation default** chosen by whoever runs the server. The page always tells you what that default currently is.

NOTE

Changing a setting takes effect immediately for everyone. Each change is written to the audit log as **settings.updated** with the names of the settings that changed - never their values, so a secret is never recorded.

Branding

Setting	What it does
Display name	The name shown on the sign-in page and in the top-left of the portal. Blank uses your organisation name.
Accent colour	The colour of buttons, links and highlights. Choose one with enough contrast against white; Use the default puts it back.
Logo	A small image shown above the sign-in form and beside the name in the header. Keep it under 300 KB; a wide image of roughly 200 by 48 pixels works best. Remove the logo clears it.

Branding is what tells people they are in the right place, so use the name your colleagues recognise rather than a legal entity name.

Who can register

Nobody is given an account by an administrator: people create their own at first sign-in (chapter 7). These two settings decide who is allowed to.

Setting	What it does
Allowed email domains	A comma-separated list, for example acme.com, acme.co.uk. Only addresses at those domains may register. Blank means any address the installation allows.
Email verification at first sign-in	On means a one-time code is emailed and must be typed in before a password key can be created. Off means anyone who knows an address that has no key yet can claim it.

NOTE

Leave verification **on** unless you have a specific reason not to. With it off, somebody who knows a colleague email address can take that account before its owner ever signs in. Turning it off does not remove the need for working email either, because a forgotten key always uses an emailed code.

Restricting the domains does not affect people who already have an account, and it does not stop an **approver chosen from a spreadsheet** being given one: that person is created by the workflow, not by registration.

Email sender

The **From name** and **From address** your notifications are sent as - approval requests, reminders, decisions and verification codes. The page shows the address currently in use. Ask your IT team before changing it: a mail server normally refuses to send as an address it is not authorised for, and messages that do go out may be treated as spam. The mail server itself is an installation-wide setting and is not on this page.

CHAPTER 13

Common tasks and troubleshooting

Situation	What to do
An approver has left	Requests > filter Approver > Reassign each open step. Forms > publish a new chain without them. Users > Deactivate (this also stops the address being registered again).
An approver is on leave	Add a delegate on their open steps (Reassign... with the box ticked). For long absences publish a chain naming the stand-in, and switch back afterwards.
Approver says the link does not work	Check they sign in with the right account. Send reminder issues a fresh link. They can also use "Waiting for my approval" on their home page.
Approver did not get the email	Open the request > Emails : if Sent, ask them to check junk mail; if Failed, fix the cause and Resend ; then Send reminder .
User forgot their password key	Point them to Forgot your key? on the sign-in page. If they cannot receive the email: Users > Reset password key .
User cannot sign in	Locked: wait 15 minutes, or they use Forgot your key? (which unlocks). Inactive: Reactivate . Verification code not arriving: ask them to check junk mail and to request a new code after a minute; if nobody receives codes, the mail server needs attention from IT.
Someone needs to approve but is not in the approver list	They must sign in once; then tick Approver for them on the Users page.
Request submitted by mistake	Cancel request... with a reason. The submitter can submit again.
Submitter wants a rejection reversed	Not possible. They start a new request; "start a new request using these details" on the rejected request pre-fills it.
Wrong approver on in-flight requests after a chain change	Expected: in-flight requests keep their original chain. Reassign the affected steps.
Form is missing for submitters	It must be Active and have a published chain with active approvers.
"The approver for step ... is inactive" on submit	Publish a new chain version with an active approver.
Need evidence of who did what	Audit log > filter > Export CSV ; and the request's PDF, which includes its audit summary.

Security reminders

- Give the Admin role sparingly, and review the user list regularly. Deactivate leavers the same day.
- Never ask users for their password key or verification code; you do not need them for anything. To help someone who is stuck, use **Reset password key**.
- A 6-digit key is short. The protections are the 15-minute lock after 5 wrong attempts and the emailed verification code at setup - do not ask IT to switch the verification off without understanding that anyone could then claim a colleague's address.
- Approval links are personal and require sign-in, so a forwarded email cannot be used by someone else - but treat the audit entry **auth.refresh_reuse_detected** and unexpected **auth.login_failed** bursts as worth a look.
- Exports and PDFs contain personal and business data. Store them according to your organisation's policy.